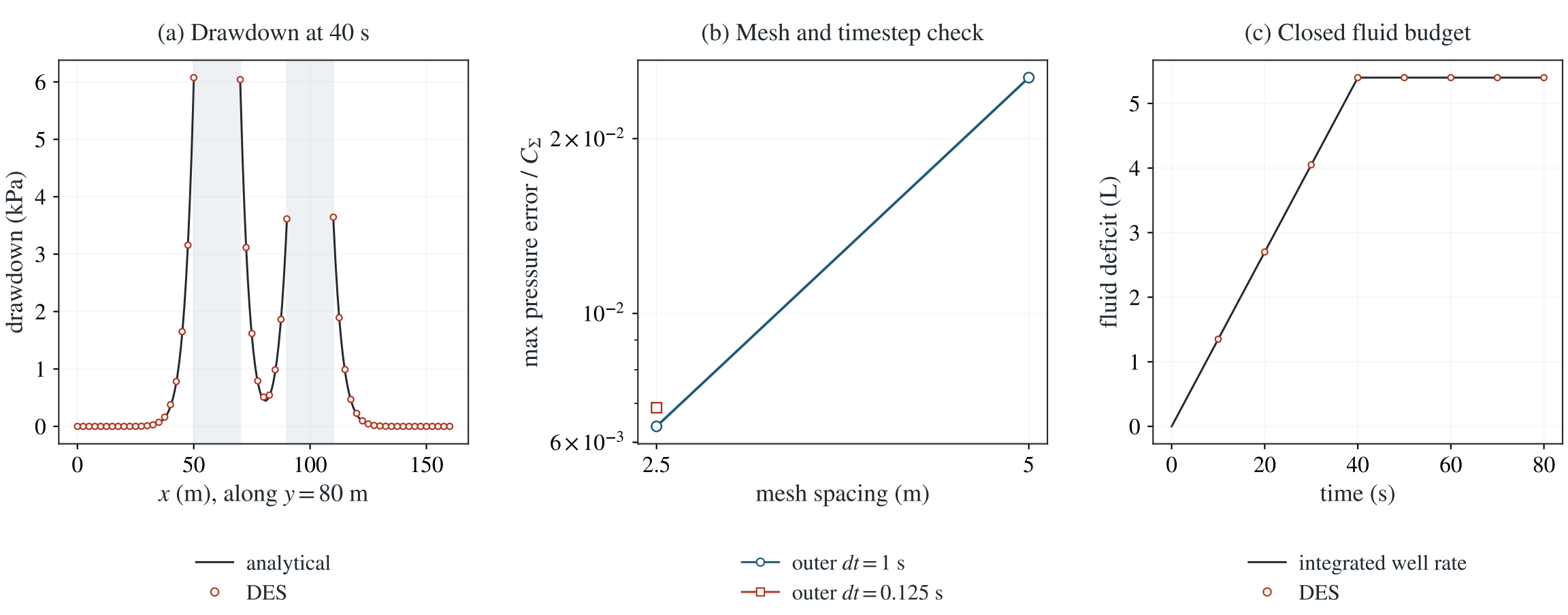

(a) Drawdown at 40 s
drawdown (kPa)
0
1
2
3
4
5
6
0
50
100
150
x (m), along y = 80 m
analytical
DES
(b) Mesh and timestep check
max pressure error / C_Σ
2×10^{-2}
10^{-2}
6×10^{-3}
2.5
5
mesh spacing (m)
outer $dt = 1$ s
outer $dt = 0.125$ s
(c) Closed fluid budget
fluid deficit (L)
0
1
2
3
4
5
0
20
40
60
80
time (s)
integrated well rate
DES